Supplementary Material

Capacity-curve surrogate for planar reinforced concrete frames: accuracy and preservation of the stirrup-spacing effect

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This supplement gives the details of the frame population and the numerical model (Section S1), the checks of the spacing comparison (Section S2), the surrogate (Section S3) and the single-option re-analysis (Section S4). In the tables and figures, P3 denotes the main population and P0 the sensitivity population. The frame data, the trained models and the specimen list of the column tests are available in the repository given in the Data availability statement of the main text; the pre-registration records and the analysis scripts are available from the authors on request.

S1. Frame population and numerical model

The frames were generated by Latin hypercube sampling in each story stratum (two to six stories). Each stratum had a quota of 5,840 accepted frames in P3 and 2,280 in P0. If a quota was not filled, the stratum was resampled with a doubled sample size. Columns are square, with the side rounded to 0.05 m and at least 0.30 m. The filters were applied in the following order: beam flexural bars fitting in one layer, longitudinal bars fitting in the section, an approximate strong-column condition at the most critical interior joint ($\sum M_c \geq 1.2 \sum M_b$), a gravity axial-load ratio below 0.60 and the TS 500 static beam-shear check. Of the 222,883 P3 candidates, 23,784, 14,982, 154,788 and 129 were rejected by these filters in turn, and none by the axial-load filter. This left 29,200 frames (13.1%). Of the 13,130 P0 candidates, 1,078, 100, 443 and 109 were rejected, which left 11,400 frames. P0 draws the target axial-load ratio from $U[0.10,0.40]$ and enlarges the column wherever the beam is deeper than the column side (92.3% of its frames).

Stirrup diameter, spacing and column legs were independent sampling dimensions, with a single spacing along each member. The largest absolute Spearman correlation among them was 0.010 in P3 and 0.020 in P0. The frame-specific confinement-zone spacing limits of TBEC 2018 (7.3.4.1(a), 7.4.4) ranged from 8.4 to 13.2 cm in columns and from 9.6 to 15.0 cm in beams (10th to 90th percentiles). Their medians were 10.8 and 12.8 cm, and 41.4% (columns) and 50.6% (beams) of the training-pool frames met them. Table S1 lists the variables, the filters and the TBEC 2018 rules applied, approximated or not applied.

Table S1. Sampled frame population: sampling filters, TBEC 2018 rules applied or not applied, and frame-specific spacing limits, for the main population (P3) and the sensitivity population (P0). The sampled variables are listed in Table 2 of the main text.

(a) Sampling filters, applied in this order (rejected candidates)

Filter	**P3**	**P0**
Beam flexural bars required by TS 500 fit in one layer	23,784	1,078
Longitudinal bars feasible (25 mm clear spacing; column ρ_c 0.01–0.04)	14,982	100
Strong column: $\Sigma M_c \geq 1.2\ \Sigma M_b$ at interior joints (characteristic strengths, gravity axial load)	154,788	443
Beam shear reinforcement, TS 500 static check ($1.4G + 1.6Q$)	129	109
Accepted / candidates processed	29,200 / 222,883	11,400 / 13,130

(b) TBEC 2018 rules for high-ductility frames in the population design (training pool, n = 20,432)

Clause	Requirement	Status	Check in the training pool
7.3.1.1, 7.4.1.1	Minimum member dimensions	Applied (filter)	no violation
7.3.1.2	$N_{dm} \le 0.40\, A_c f_{ck}$	Approximate: gravity axial load only, filter limit 0.60	2 frames above 0.40
7.3.2.1	Column longitudinal ratio 0.01–0.04; bars ≥ φ14	Applied (sampling range)	no violation
7.4.2.1–7.4.2.4	Beam longitudinal reinforcement limits	Applied (sampling and filter)	no violation
7.3.4.1, 7.4.4	Confinement-zone stirrup spacing; column A_{sh} (Eq. 7.1)	Not applied: sampled across the limit	part (c); column spacing and A_{sh} both met: 14.4%
7.3.5	Strong column	Approximate (filter in part (a))	–
7.3.7, 7.4.5	Shear design from moment capacities	Not applied (beams: TS 500 static check only)	–
7.5.2	Joint shear	Not applied	Section S1
Chapter 4	Earthquake design for a spectrum	Not applied: frames sampled, not designed	–

(c) Frame-specific TBEC spacing limits in the confinement zones (training pool)

Element	Clause	Limit	P10 (cm)	Median (cm)	P90 (cm)	Frames within the limit (%)
Column	7.3.4.1(a)	min(b_{min}/3, 150 mm, 6φl)	8.4	10.8	13.2	41.4
Beam	7.4.4	min(d/4, 8φl, 150 mm)	9.6	12.8	15.0	50.6

Notes. d, effective beam depth; φl, longitudinal bar diameter; b_{min}, smaller column dimension.

The elastic interior of each force-based element has the effective stiffness that TBEC 2018 (5.4.5.2) gives for lumped models,

$$EI_e = M_y L_s / (3\,\theta_y) \le EI_g \tag{S1}$$

where M_y and θ_y come from a moment–curvature analysis of the same fiber section under its gravity axial force, EI_g is the gross stiffness, and L_s is half the story height in columns and half the span in beams. The fiber end regions have the code plastic-region length $L_p = 0.5h$ (5.3.1.2). This length was limited so that the interior weight of the modified Gauss–Radau rule, whose integration points span $4L_p$ per end [1], stays non-negative:

$$L_{p,e} = \min(0.5h,\, L_{fl}/8) \tag{S2}$$

where L_{fl} is the flexible length between the rigid end zones. The limit shortened the plastic region of 37.8% of the P3 column elements (in 41.7% of the frames) and of 0.9% of the beam elements. In P0, it shortened 27.0% of the column elements. Rigid end zones are half the column width at the beam ends and half the beam depth at the column ends, with none at the column bases. Bar centers lie at the cover plus the stirrup diameter plus half the bar diameter from the face, and the confined core extends to the stirrup centerline. The beam load $G + 0.3Q$ acts on the flexible length. The load on the rigid zones is applied as an equivalent nodal force and moment. The column self-weight (25 kN/m³) is a nodal load and story mass, and the concrete tensile strength is $0.10\, f_c$. The confined core crushes at $\varepsilon_{cu} = 0.0035 + 1.4\rho_s f_{yw} \varepsilon_{su} / f_{cc}$, the relation of Priestley et al. [2] with the base term taken as 0.0035 so that

the crushing strain is consistent with the TBEC 2018 concrete strain limits (Eq. (1) of the main text). This choice is a model assumption and was not calibrated against tests.

No run was excluded. Numerical termination, an end through loss of convergence or a rejected equilibrium step before any bar ruptured or reached $0.9\varepsilon_{\mathrm{su}}$, occurred in 15.6% of the P3 and 16.9% of the P0 frames. Numerical variability limits the interpretation of comparisons between single runs. In a test with negligibly perturbed inputs, changes exceeding 3% occurred in 8.2% of the comparisons (12 of 147 perturbed runs) for both the column LS and CP drifts. Joint deformation was not modeled. In a post hoc check, a joint exceeded the code shear limit in 18.6% of the training-pool frames when the joints were taken as unconfined and in 0.2% when they were taken as confined.

The shear capacity of Eq. (2) of the main text uses the TS 500 [3] concrete contribution and the TBEC 2018 upper bound:

$$V_c = 0.8 \times 0.65\, f_{\mathrm{cte}} b_w d\,(1 + 0.07N/A_c),\ f_{\mathrm{cte}} = 0.35\sqrt{f_{\mathrm{ce}}} \tag{S3}$$

$$V_{\mathrm{ub}} = 0.85\sqrt{f_{\mathrm{ce}}}\, A_w \tag{S4}$$

where N is the axial force and A_c the gross section area. $A_w = bh$ in columns (TBEC 2018 Eq. (7.7), 7.3.7.5) and $A_w = b_w d$ in beams (Eq. (7.10), 7.4.5.2). Expected strengths were used with unit material factors. The concrete contribution is set to zero in beams whose seismic shear, the end shear minus its gravity part, exceeds half of the total shear (7.4.5.3). In columns, it is set to zero if, in addition, $N \leq 0.05 A_c f_{\mathrm{ck}}$ (7.3.7.6).

Table S2 compares the production setting (effective stiffness and expected strengths) with gross stiffness and characteristic strengths on a benchmark of 50 sampled frames. The absolute event-to-demand ratios depend on this setting. For this reason, the main text interprets group differences rather than absolute shares.

Table S2. Model setting of the frame analyses on a benchmark of 50 sampled frames: (a) pre-registered comparison of four settings (default element formulation, raw output) and (b) the production setting against gross stiffness and characteristic strengths (element compatibility iteration, cleaned curves, 22 frames accepted in all four settings).

(a) Pre-registered comparison, default element formulation, raw output

Quantity	Statistic	B	R	D	RD (production)
T_1	Ratio to B per frame, median (P10–P90); n	1 (reference)	1.95 (1.74–2.14); n = 34	0.94 (0.88–0.96); n = 49	1.85 (1.64–1.98); n = 33
δ_t, 2475-yr ZC	Ratio to B per frame, median (P10–P90); n	1 (reference)	2.01 (1.80–2.22); n = 34	0.93 (0.86–0.96); n = 49	1.90 (1.71–2.14); n = 33
V_{max}	Ratio to B per frame, median (P10–P90); n	1 (reference)	0.95 (0.87–0.98); n = 34	1.19 (1.17–1.20); n = 49	1.09 (0.62–1.17); n = 33
$\delta_{LS,b}$ / δ_t, 2475-yr ZC	KM median (censored frames); n	1.15 (0); n = 24	0.51 (0); n = 24	1.21 (0); n = 24	0.56 (0); n = 24
$\delta_{LS,c}$ / δ_t, 2475-yr ZC	KM median (censored frames); n	1.44 (1); n = 24	1.13 (1); n = 24	1.58 (0); n = 24	1.28 (2); n = 24

(b) Production setting RD against B, element compatibility iteration, cleaned curves

Quantity	B, median (P10–P90)	RD, median (P10–P90)	RD/B	n
T_1 (s)	0.39 (0.21–0.73)	0.73 (0.38–1.33)	1.83 (1.64–1.96)	22
δ_t, 2475-yr ZC (m)	0.116 (0.050–0.215)	0.216 (0.106–0.384)	1.87 (1.72–2.11)	22
V_{max} (kN)	881 (458–1462)	898 (476–1671)	1.11 (0.79–1.17)	22
$\delta_{LS,b}$ / δ_t, 2475-yr ZC (KM median; censored)	1.13 (0)	0.57 (0)	0.51	22
$\delta_{LS,c}$ / δ_t, 2475-yr ZC (KM median; censored)	1.54 (0)	1.37 (3)	0.89	22

Notes. B, gross stiffness and characteristic strengths; R, effective stiffness (Eq. (S1)); D, expected strengths; RD, both (production setting); n, frames accepted in B and in the setting; KM, Kaplan–Meier.

The column model was compared with 73 column tests of the PEER Structural Performance Database [4] (Table S3). The specimen list and the original test sources are given in the repository. The comparison concerns the monotonic response and the drifts of the code damage events. The tests do not show whether the model reproduces the effect of stirrup spacing. A pre-registered spacing-trend test on six specimens was inconclusive in both the tests and the model, and the specimen set was fixed after the first results.

Table S3. Column model against 73 rectangular reinforced concrete column tests of the PEER Structural Performance Database [4]: (a) model-to-test ratios of the monotonic response; (b) model IO and CP event drifts against the observed cover spalling and bar buckling [5]; (c) pre-registered spacing-trend test on the six specimens of one test series.

(a) Model-to-test ratios, cantilever models of the 73 specimens

Quantity (model/test)	Gross: n	Gross: mean	Gross: median	Gross: COV	Gross: range	R: n	R: mean	R: median	R: COV	R: range
Peak force V_{peak}	73	0.94	0.92	0.13	0.76–1.23	73	0.94	0.92	0.13	0.76–1.23
Initial stiffness K	73	3.89	2.79	0.72	1.37–13.58	73	1.49	1.10	0.64	0.62–4.69
Yield drift θ_y	73	0.32	0.31	0.43	0.08–0.68	73	0.79	0.78	0.39	0.23–1.48
Drift at peak θ_{peak}	73	0.93	0.72	0.78	0.19–3.82	73	1.14	0.92	0.66	0.26–4.13
Drift at 20% strength loss $\theta_{0.8}$	40	1.02	1.06	0.34	0.41–1.66	37	1.10	1.11	0.31	0.51–1.74

(b) Model events against observed damage

Comparison	Stiffness	Matched specimens	Model censored	Median ratio	Geometric mean	Log standard deviation	Ratio in [0.5, 2] (%)	Ratio < 1 (%)	Model event governed by concrete (%)
Model IO / test cover spalling	Gross	39	0	0.35	0.37	0.46	25.6	97.4	87.2
Model IO / test cover spalling	R	39	0	0.65	0.72	0.45	74.4	74.4	87.2
Model CP / test bar buckling	Gross	21	0	0.44	0.48	0.34	38.1	100.0	85.7
Model CP / test bar buckling	R	21	0	0.56	0.59	0.33	61.9	95.2	85.7
Model cover strain 0.004 / test cover spalling (supplementary)	Gross	39	0	0.48	0.45	0.43	46.2	97.4	—
Model cover strain 0.004 / test cover spalling (supplementary)	R	39	0	0.80	0.81	0.43	79.5	66.7	—

(c) Spacing-trend test, the six specimens of the Pujol 10 series of the database

Response	Source	Element iteration (iterations; tolerance)	Slope b (%/mm)	95% CI	CI includes 0	R^2	Censored specimens (of 6)
$\theta_{0.8}$ (primary test)	Test	—	−0.0220	[−0.0815, 0.0375]	yes	0.66	0
$\theta_{0.8}$ (primary test)	Model, gross	default (10; 10^{-12})	−0.0299	[−0.0994, 0.0395]	yes	0.67	3
$\theta_{0.8}$ (primary test)	Model, R	default (10; 10^{-12})	−0.0696	[−0.3111, 0.1719]	yes	0.48	5
$\theta_{0.8}$ (primary test)	Model, gross	200; 10^{-8}	0.0064	[−0.0546, 0.0675]	yes	0.59	0
$\theta_{0.8}$ (primary test)	Model, R	200; 10^{-8}	0.0059	[−0.0206, 0.0324]	yes	0.79	0
θ_{peak} (descriptive)	Test	—	0.0009	[−0.0052, 0.0071]	yes	0.99	—
θ_{peak} (descriptive)	Model, gross	default (10; 10^{-12})	−0.0244	[−0.0817, 0.0330]	yes	0.77	—
θ_{peak} (descriptive)	Model, R	default (10; 10^{-12})	−0.0587	[−0.2732, 0.1559]	yes	0.45	—
θ_{peak} (descriptive)	Model, gross	200; 10^{-8}	0.0077	[−0.0401, 0.0555]	yes	0.63	—
θ_{peak} (descriptive)	Model, R	200; 10^{-8}	0.0076	[−0.0418, 0.0570]	yes	0.63	—

Notes. Specimens: flexure-failure rectangular columns with shear span to depth ratio of at least 2, concrete strength 20–45 MPa and axial-load ratio up to 0.40 (85 database records; three gave no test envelope, and the N and S records of nine test assemblies count as one specimen each, leaving 73 specimens). Gross and R, interior region with gross stiffness and with the effective stiffness of Eq. (S1); COV, coefficient of variation; $\theta_{0.8}$, drift at which the force falls to $0.8V_{peak}$ after the peak. In (b), ratio = model event drift / test drift at the observed damage. In (c), b is the slope of the drift against the stirrup spacing.

S2. Checks of the spacing comparison

Five pre-registered variants were compared with the primary Kaplan–Meier estimate (Table S4 for P3, with P0 in the repository). The first is Aalen–Johansen cumulative incidence with the first bar rupture as a competing event [6]. The second is Kaplan–Meier stratified by story count, and the third is the raw rule, in which events after the first rupture are counted. The last two variants exclude flagged runs and early-rupture runs. A run is flagged when its first beam rupture follows the beam IO event within $0.0005H$ (286 P3 and 195 P0 frames). A run has an early rupture when its first rupture occurs below $0.01H$ (580 P3 and 325 P0 frames).

Table S4. Estimator and data-rule sensitivity of the spacing effect on the first column and beam strain events: change of the 10th percentile (P10) and the median from the 5–8 to the 17–20 cm group. Main population (P3); the sensitivity population is given in the repository.

Event (grouping)	Estimator or data rule	n, 5–8 / 17–20 cm	P10 change (%) [95% CI]	Median change (%) [95% CI]
Column LS	Kaplan–Meier, primary rule	5,080 / 5,162	−12.78 [−13.95, −11.29]	−12.05 [−13.38, −10.78]
Column LS	Aalen–Johansen (rupture as competing event)	5,080 / 5,162	−12.60 [−13.96, −11.25]	−12.30 [−13.45, −11.16]
Column LS	Kaplan–Meier stratified by story count	5,080 / 5,162	−12.79 [−13.92, −11.44]	−12.06 [−13.35, −10.98]
Column LS	Raw rule (events after rupture counted)	5,080 / 5,162	−12.71 [−13.93, −11.20]	−11.96 [−13.28, −10.81]
Column LS	Flagged runs excluded	5,011 / 5,084	−12.78 [−13.86, −11.30]	−12.05 [−13.29, −10.81]
Column LS	Early rupture (< 0.01H) excluded	4,935 / 5,010	−12.73 [−13.87, −11.31]	−12.01 [−13.39, −10.81]

Event (grouping)	Estimator or data rule	n, 5–8 / 17–20 cm	P10 change (%) [95% CI]	Median change (%) [95% CI]
Column CP	Kaplan–Meier, primary rule	5,080 / 5,162	−17.26 [−18.46, −15.66]	−15.73 [−16.90, −14.55]
Column CP	Aalen–Johansen (rupture as competing event)	5,080 / 5,162	−17.37 [−18.45, −15.53]	−16.30 [−17.67, −15.11]
Column CP	Kaplan–Meier stratified by story count	5,080 / 5,162	−17.35 [−18.57, −15.61]	−15.88 [−17.01, −14.70]
Column CP	Raw rule (events after rupture counted)	5,080 / 5,162	−17.20 [−18.60, −15.64]	−15.47 [−16.59, −14.15]
Column CP	Flagged runs excluded	5,011 / 5,084	−17.26 [−18.56, −15.66]	−15.73 [−16.80, −14.53]
Column CP	Early rupture (< 0.01H) excluded	4,935 / 5,010	−17.24 [−18.43, −15.50]	−15.72 [−16.94, −14.55]
Beam LS	Kaplan–Meier, primary rule	5,080 / 5,093	−0.86 [−2.12, +0.49]	+0.20 [−0.79, +1.17]
Beam LS	Aalen–Johansen (rupture as competing event)	5,080 / 5,093	−0.86 [−2.19, +0.38]	+0.20 [−0.87, +1.10]
Beam LS	Kaplan–Meier stratified by story count	5,080 / 5,093	−0.99 [−2.24, +0.29]	−0.02 [−0.70, +0.90]
Beam LS	Raw rule (events after rupture counted)	5,080 / 5,093	−0.86 [−2.17, +0.49]	+0.20 [−0.85, +1.25]
Beam LS	Flagged runs excluded	5,036 / 5,003	+0.40 [−0.71, +1.43]	+0.44 [−0.55, +1.60]
Beam LS	Early rupture (< 0.01H) excluded	4,964 / 4,931	+0.47 [−0.64, +1.26]	+0.59 [−0.49, +1.69]
Beam CP	Kaplan–Meier, primary rule	5,080 / 5,093	−0.89 [−1.97, +0.61]	−0.02 [−1.10, +0.94]
Beam CP	Aalen–Johansen (rupture as competing event)	5,080 / 5,093	−0.89 [−2.06, +0.65]	−0.02 [−1.06, +0.99]
Beam CP	Kaplan–Meier stratified by story count	5,080 / 5,093	−1.08 [−2.09, +0.49]	−0.17 [−0.94, +0.61]
Beam CP	Raw rule (events after rupture counted)	5,080 / 5,093	−0.89 [−2.04, +0.60]	−0.02 [−1.05, +0.91]
Beam CP	Flagged runs excluded	5,036 / 5,003	+0.33 [−0.71, +1.59]	+0.15 [−0.96, +1.22]
Beam CP	Early rupture (< 0.01H) excluded	4,964 / 4,931	+0.39 [−0.56, +1.22]	+0.06 [−0.92, +1.27]

Notes. Column events are grouped by column spacing and beam events by beam spacing. The variants are described in Section S2. CIs: 95% bootstrap intervals.

The post hoc censoring check (Fig. S1) places the event of every censored frame either at infinity or at its censoring drift. The same assumption is applied in both spacing groups, or opposite assumptions are applied to the two groups. The opposite assumptions give assumption bounds, not confidence intervals.

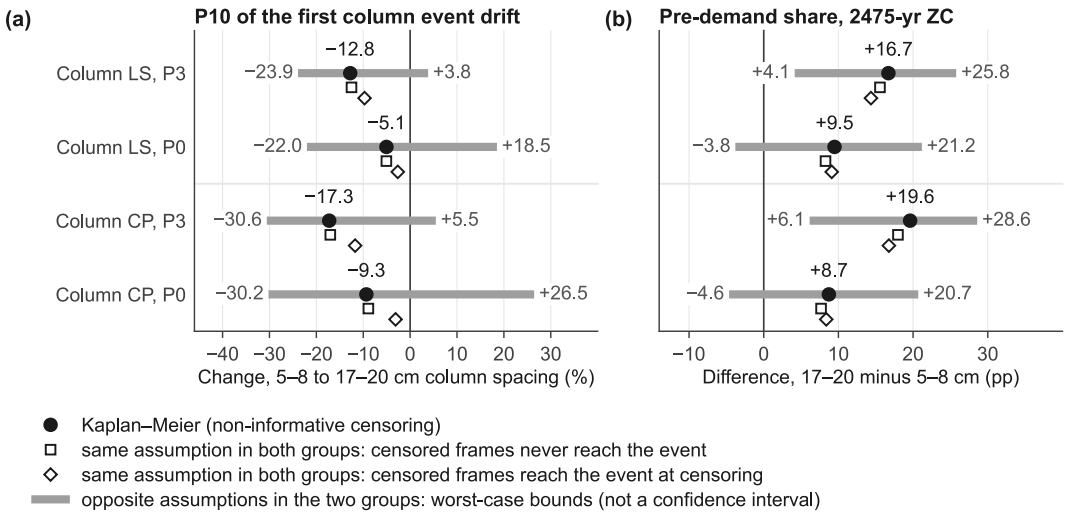


Fig. S1. Censoring sensitivity of the column-spacing contrasts (post hoc). (a) Change of the 10th percentile of the column LS and CP drifts from the 5–8 to the 17–20 cm group. (b) Difference in the share of frames whose column LS or CP event precedes the 2475-year target displacement on soil class ZC. Kaplan–Meier estimate and assumption bounds, P3 and P0.

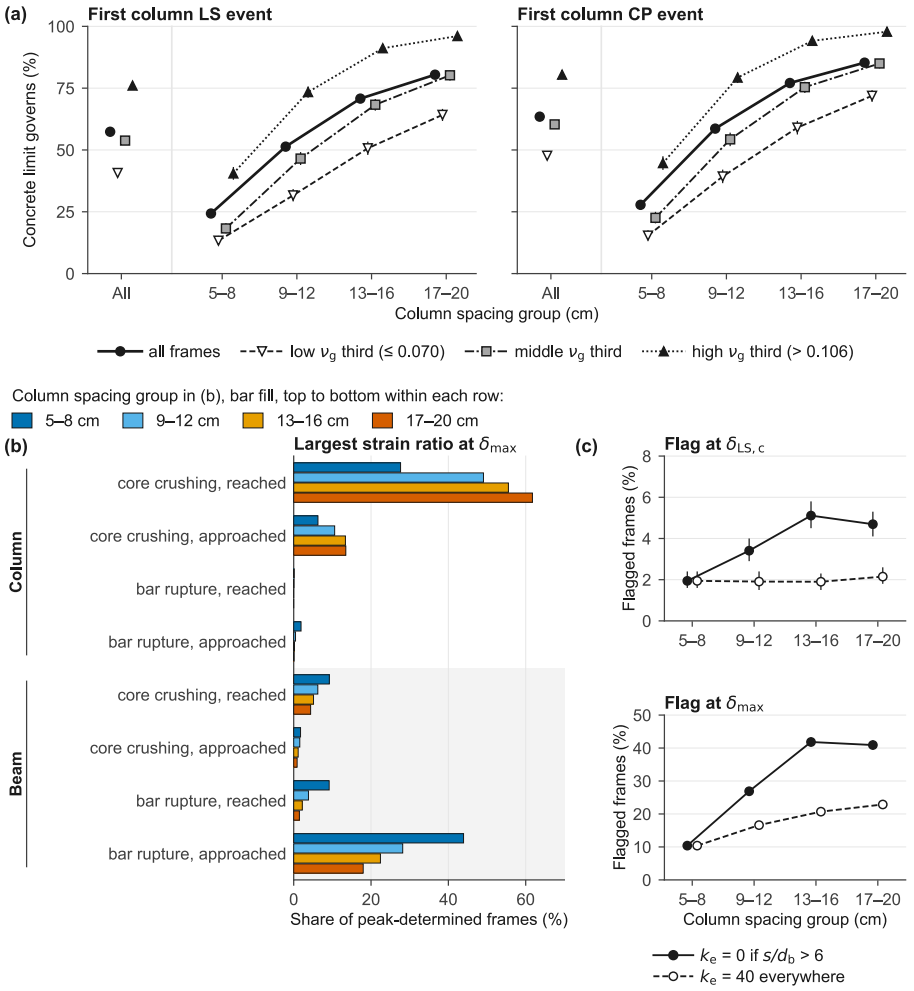


Fig. S2. Column events in P3. (a) Share of the first column LS and CP events governed by the concrete strain limit, by spacing group and axial-load third (post hoc). (b) Element and limit with the largest strain ratio at the peak base shear. (c) Share of frames with a bar-buckling flag.

Fig. S2 shows which limit governed the first column events and how often the bar-buckling flag was raised. Fig. S3 gives the shear-design classes of the beams. It supports the statement in Section 3.2 of the main text that beams meeting the capacity-shear requirement showed almost no shear exceedance.

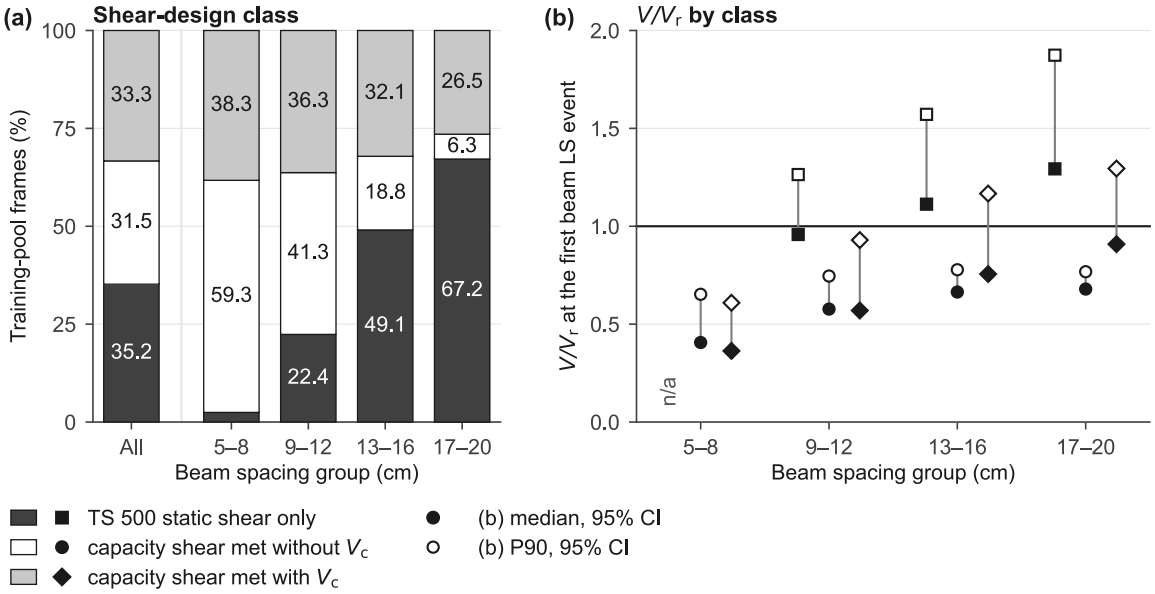


Fig. S3. Beam shear-design classes in P3 (post hoc). (a) Share of the classes by beam spacing group. (b) Beam V/V_r at the first beam LS event by class and spacing group.

S3. Surrogate

The 34 inputs comprise geometry (11), materials (5), loads (5), longitudinal reinforcement (8) and stirrups (5). Targets were learned on a log scale, and errors are computed on the original scale. Cross-validation was five-fold at frame level and stratified by the 80 cells of the hold-out split, with the same folds for all families. Table S5 gives the family-selection rule with its scores and the selected hyperparameters. Fig. S4 shows the curve error, the spread of the error over the cells of the hold-out split and the deviation by spacing group. Table S5(c) compares, at group level, the quantiles of the uncensored predictions of the primary model and of a censoring-aware model (SurvivalAft) with the Kaplan–Meier quantiles of the column events. The differences combine the prediction error and the different treatment of censored frames.

The hold-out set was evaluated after each of three successive versions of the database, the first two of which were superseded by corrections of the numerical model. In each version, the hyperparameters were tuned by cross-validation before the hold-out evaluation and were not changed afterward, the pre-registered rule selected CatBoost, and the definition of the preservation test was unchanged. One change was informed by the hold-out set. The participation factor and the effective mass ratio were added as targets after an error analysis on hold-out predictions. The out-of-fold predictions showed the same error. These targets enter only the derived target displacement.

Table S5. Surrogate training details: (a) pre-registered model-family selection; (b) selected hyperparameters; (c) group-level comparison of predicted column-event quantiles with Kaplan–Meier quantiles.

(a) Model-family selection (documentation of a step of the method, not a method comparison)

Row	CatBoost	Ridge	Random Forest
CV MdAPE (%) and rank, T_1	0.81 (1)	4.56 (2)	4.95 (3)
CV MdAPE (%) and rank, V_y	2.72 (1)	8.27 (2)	9.89 (3)
CV MdAPE (%) and rank, δ_y/H	4.03 (1)	7.68 (2)	7.86 (3)
CV MdAPE (%) and rank, V_{max}	2.48 (1)	8.05 (2)	9.94 (3)
Rank mean over the four targets	1.00	2.00	3.00
Mean CV MdAPE over the four targets (%)	2.51	7.14	8.16
Tuning trials per target	60	41	40
Selected	yes	no	no

(b) Selected CatBoost hyperparameters

Target	Training frames	Depth	Learning rate	L2 leaf regularization	Bagging temperature	Trials	CV criterion (%)	Folds with best iteration ≥ 2,900	Final model, best iteration
T_1	20,432	6	0.0539	1.231	0.777	60	0.81	5/5	2,998
V_y	12,213	7	0.0234	1.428	0.612	60	2.72	2/5	2,749
δ_y/H	12,213	8	0.0260	1.070	0.634	60	4.03	3/5	2,996
V_{max}	12,213	7	0.0394	1.321	0.829	60	2.48	2/5	2,009
$\delta_{IO,b}$	20,260	8	0.0409	1.668	0.314	60	2.30	4/5	2,988
$\delta_{LS,b}$	19,805	10	0.0281	1.081	0.814	60	3.06	2/5	1,287
$\delta_{CP,b}$	19,505	10	0.0204	1.034	0.794	60	3.71	1/5	1,730
$\delta_{IO,c}$	18,566	6	0.0616	2.441	0.923	60	1.66	5/5	2,999
$\delta_{LS,c}$	17,015	6	0.0808	4.001	0.834	60	2.07	5/5	2,999
$\delta_{CP,c}$	15,861	6	0.0754	11.658	0.933	60	2.36	5/5	2,997
V/V_r at $\delta_{LS,b}$	19,805	9	0.0203	1.101	0.750	60	4.45	1/5	2,997
Γ_1	20,432	7	0.0516	2.649	0.804	60	0.09	5/5	2,999
M_1^*/M	20,432	6	0.0755	4.469	0.746	60	0.10	5/5	2,998
Pushover curve (primary / all points)	20,432	4	0.1098	1.932	0.259	20	1.42 (1.81 on a 30% subset)	5/5	2,999 / 2,794

Notes. CV MdAPE, mean of the five fold errors of the cross-validation on the training pool. (a) The family with the lowest mean rank over T_1, V_y, δ_y/H and V_{max} was selected, by a rule fixed before the hold-out set was opened. Tuning: Ridge over 41 penalty values; Random Forest and CatBoost with a tree-structured Parzen estimator [7]. (b) Iteration limit 3,000 with early stopping; the final models were refitted on the whole training pool.

(c) Predicted column-event quantiles against Kaplan–Meier quantiles (training pool, out-of-fold predictions)

Event	Model	Censored frames with prediction < k_d (%)	P10 vs KM, by column group (%)	Median vs KM, by column group (%)	P10 / median vs KM, all frames (%)
$\delta_{LS,c}$	Primary CatBoost (trained on reached frames)	5.1	+0.3 to +1.5	0.0 to +0.6	+0.5 / +0.2
$\delta_{LS,c}$	SurvivalAft, scale σ (primary)	5.4	−0.1 to +1.7	0.0 to +0.7	+0.4 / +0.4
$\delta_{LS,c}$	SurvivalAft, scale 2σ (robustness)	4.2	+0.2 to +1.0	−0.2 to +0.6	+0.5 / +0.3
$\delta_{CP,c}$	Primary CatBoost (trained on reached frames)	7.6	+0.4 to +1.2	−0.5 to +0.4	+0.9 / −0.1
$\delta_{CP,c}$	SurvivalAft, scale σ (primary)	6.5	0.0 to +1.4	−0.2 to +0.3	+0.1 / +0.3
$\delta_{CP,c}$	SurvivalAft, scale 2σ (robustness)	3.9	+0.2 to +1.3	0.0 to +0.6	+0.5 / +0.4

Notes (c). σ, standard deviation of the out-of-fold log residual of the primary model; k_d, drift ratio at which a censored frame leaves observation; differences are predicted quantile / Kaplan–Meier quantile − 1. A model with scale 1 failed the residual diagnostic and is not reported.

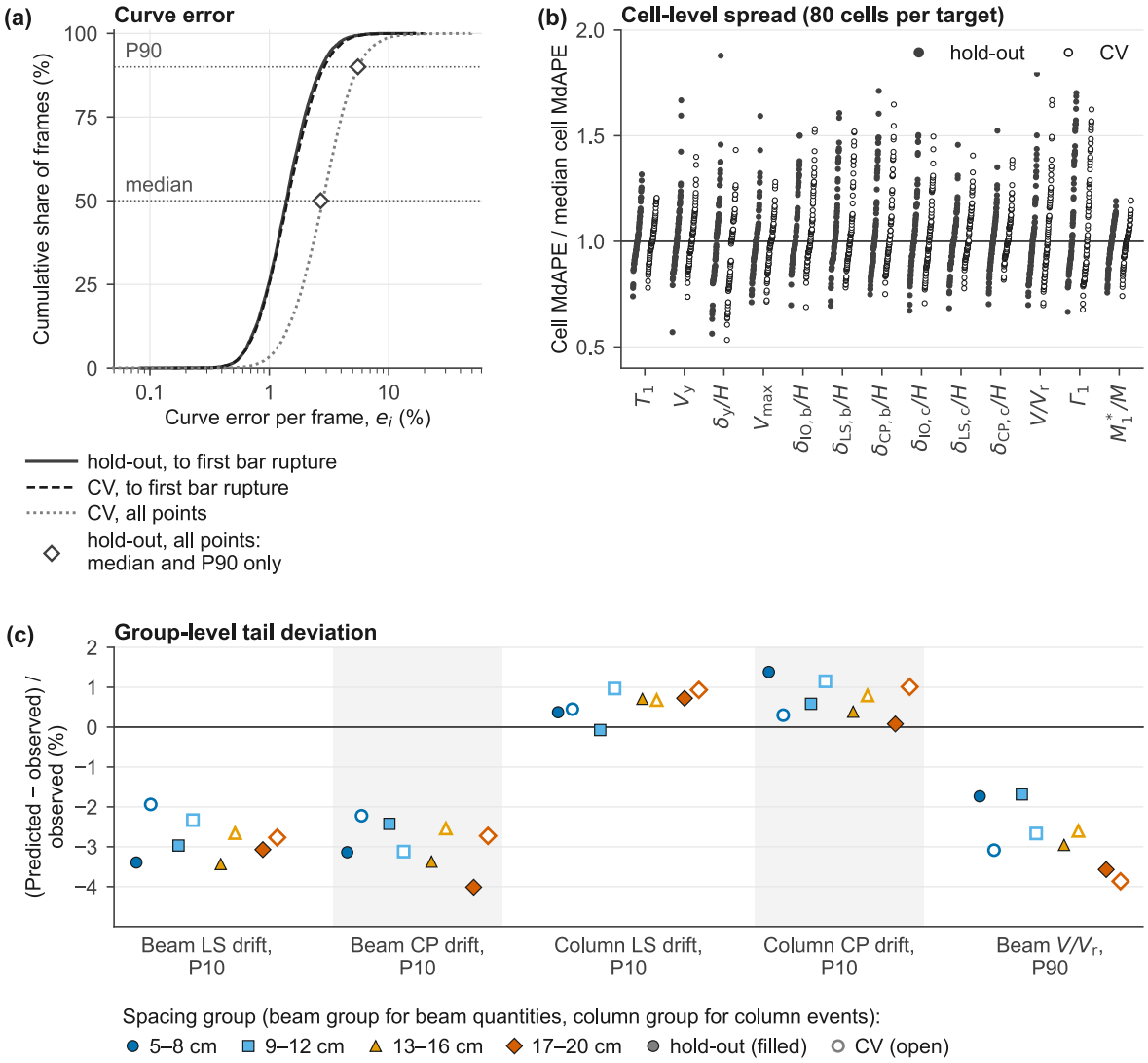


Fig. S4. Surrogate errors beyond the summary measures. (a) Curve error per frame. (b) Spread of the error over the 80 cells of the hold-out split. (c) Deviation of the predicted tail percentiles by spacing group.

S4. Single-option re-analysis

Table S6 gives the details of the single-option check in the ten re-analyzed hold-out frames: the criteria by option spacing, the recorded change of the column LS drift by the governing limit and the divergent beam LS decisions. Fig. S5 shows the first of these frames, which is also the frame of Fig. 1b of the main text, with its seven stirrup options.

Table S6. Single-option check in ten hold-out frames: details. (a) Criteria by option spacing; (b) recorded change of $\delta_{LS,c}$ by the limit governing the first column LS event; (c) divergent decisions "beam LS before δ_t".

(a) Pre-registered criteria applied to each option spacing (descriptive; the decision uses the 30 pairs of Table 4 of the main text)

Quantity	Option spacing (cm)	Pairs used (censored)	Pairs with $\|\Delta\| > 3\%$	Correct sign	Median $\|\Delta\|$ (%)	Median $\|\hat{\Delta} - \Delta\|$ (pp)	Admissible limit (pp)	Result
V/V_r at $\delta_{LS,b}$	7	10 (0)	10	10/10	45.35	0.71	22.67	met
V/V_r at $\delta_{LS,b}$	13	10 (0)	7	7/7	7.74	1.40	3.87	met
V/V_r at $\delta_{LS,b}$	20	10 (0)	10	10/10	56.16	3.71	28.08	met
$\delta_{LS,b}$	7	10 (0)	0	–	0.14	0.45	0.07	inconclusive
$\delta_{LS,b}$	13	10 (0)	0	–	0.09	0.39	0.05	inconclusive
$\delta_{LS,b}$	20	10 (0)	0	–	0.08	0.21	0.04	inconclusive
$\delta_{LS,c}$	7	8 (2)	2	2/2	1.05	1.25	0.53	inconclusive
$\delta_{LS,c}$	13	8 (2)	5	5/5	4.97	1.86	2.49	met
$\delta_{LS,c}$	20	8 (2)	7	7/7	8.83	2.13	4.41	met
$\delta_{CP,c}$	7	7 (3)	3	3/3	0.56	1.12	0.28	inconclusive
$\delta_{CP,c}$	13	8 (2)	6	6/6	7.14	2.48	3.57	met
$\delta_{CP,c}$	20	8 (2)	8	8/8	11.52	1.73	5.76	met

(b) Recorded change of $\delta_{LS,c}$ in the 24 uncensored column-option pairs (8 frames), by the limit that governed the first column LS event in the original and in the option run (post hoc)

Governing limit, original run	Governing limit, option run	Pairs	Median $\|\Delta\|$ (%)
concrete	concrete	10	6.63
concrete	steel	5	1.65
steel	concrete	4	14.45
steel	steel	5	0.25
concrete limit in at least one of the two runs		19	6.41

(c) The 17 divergent decisions "beam LS before δ_t" (of 70 runs × 10 scenarios; post hoc)

Frame	Scenario	Option	Decision, recorded / predicted	Recorded $\delta_{LS,b}/\delta_t$	Predicted $\hat{\delta}_{LS,b}/\hat{\delta}_t$
6-1	475-yr ZA	beam 7 cm	before / not before	0.334	1.271
6-1	475-yr ZA	beam 13 cm	before / not before	0.334	1.295
6-1	475-yr ZA	beam 20 cm	before / not before	0.334	1.278
6-1	475-yr ZA	column 13 cm	before / not before	0.407	1.306
6-1	475-yr ZA	original	before / not before	0.334	1.280
6-1	475-yr ZB	beam 7 cm	before / not before	0.334	1.271
6-1	475-yr ZB	beam 13 cm	before / not before	0.334	1.295
6-1	475-yr ZB	beam 20 cm	before / not before	0.334	1.278
6-1	475-yr ZB	column 13 cm	before / not before	0.407	1.306
6-1	475-yr ZB	original	before / not before	0.334	1.280
2-2	475-yr ZD	beam 7 cm	not before / before	1.018	0.982
2-2	475-yr ZD	beam 13 cm	not before / before	1.017	0.980
2-2	475-yr ZD	beam 20 cm	not before / before	1.018	0.984
2-2	475-yr ZD	column 7 cm	not before / before	1.015	0.984
2-2	475-yr ZD	column 13 cm	not before / before	1.018	0.983
2-2	475-yr ZD	column 20 cm	not before / before	1.022	0.985
2-2	475-yr ZD	original	not before / before	1.016	0.983

Notes. Frames are labeled story count–selection order (2-1 is the frame of Fig. S5). Δ and $\hat{\Delta}$, recorded and predicted change from the original detailing; pp, percentage points; admissible limit, half the median recorded $|\Delta|$ of the same pairs; inconclusive, fewer than 5 pairs with $|\Delta| > 3\%$. In (c), a ratio below 1 means that the event precedes δ_t.

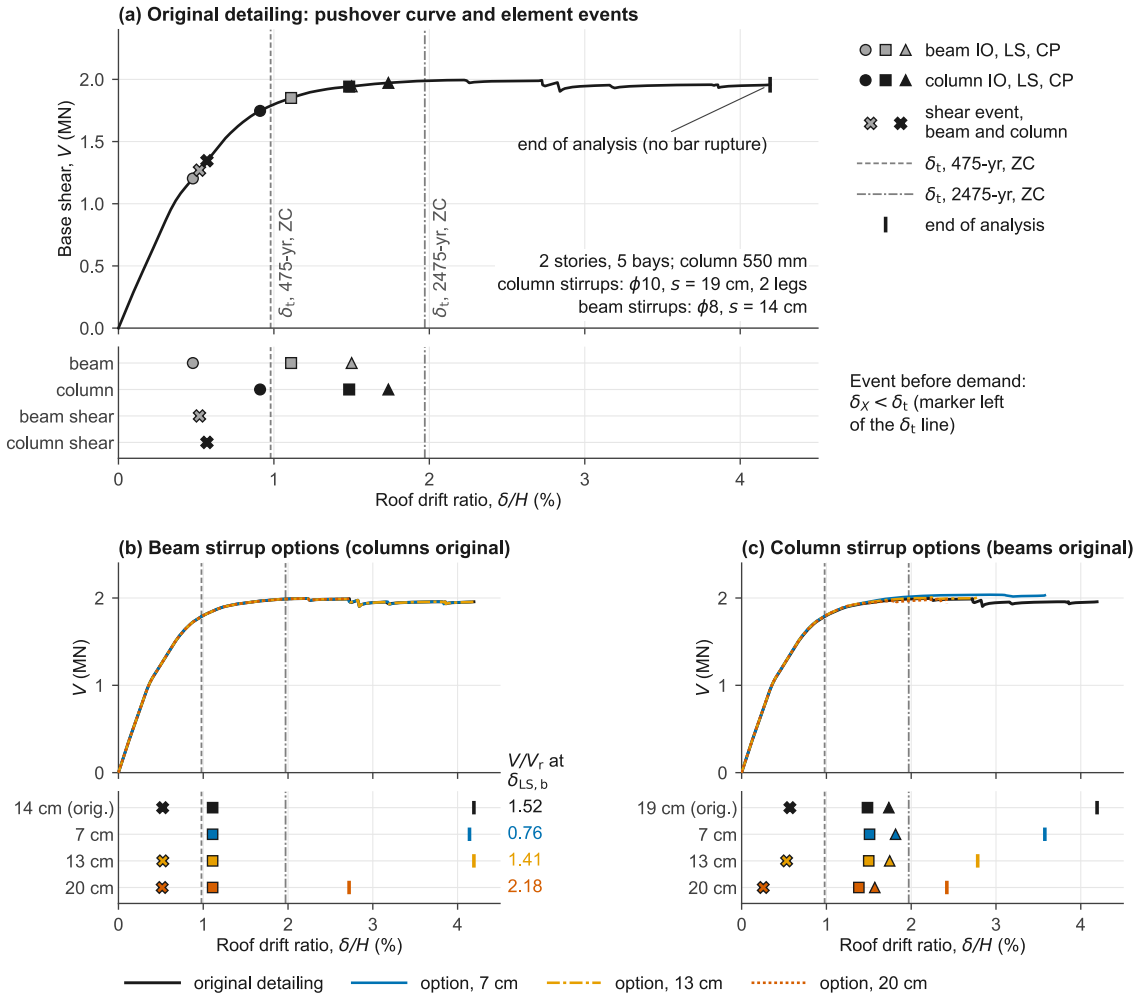


Fig. S5. Illustrative frame: the first of the ten hold-out frames selected for single-option re-analysis (two stories, five bays; column stirrups φ10 at 19 cm with two legs, beam stirrups φ8 at 14 cm). (a) Pushover curve of the original detailing with the first beam and column IO, LS and CP events, the shear events and the target displacements at the 475-year and 2475-year levels on soil class ZC. (b), (c) Beam and column stirrup options φ8 at 7, 13 and 20 cm. Strips: events of each option; a missing marker means that the event was not reached before the end of the analysis.

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